

**Department of Information Technology****Academic Year 2025– 2026 (Odd Semester)****Degree, Semester & Branch:** V Semester B.Tech. IT**Course Code & Title:** CCS334 & Big Data Analytics**Name of the Faculty member (s):** Mrs.M.Rethina Kumari**Innovative Practice Description****Unit / Topic:** Unit 4/Cassandra Hadoop Integration**Course Outcome:** CO4**Activity Chosen:** Buzz Group**Justification:**

The **Buzz Group** method promotes active participation and quick idea exchange among students in small discussion groups. It enhances analytical and creative thinking by allowing learners to evaluate integration techniques, share solutions, and collaboratively propose improvements.

Time Allotted for the Activity: 20 minutes**Details of the Implementation:**

- The class was divided into four buzz groups, each assigned a specific question related to Cassandra– Hadoop integration.
- Each group discussed and summarized their findings in 10 minutes.
- Group leaders presented their conclusions to the class in the remaining 10 minutes.
- A short reflection session was conducted to connect their ideas with real-world Big Data analytics applications.

Group	Discussion Point
Group 1	What are the advantages of integrating Cassandra with Hadoop for Big Data analytics?
Group 2	How can data be transferred between Cassandra and Hadoop? (<i>Hint: Hadoop–Cassandra Connector, Spark Cassandra Connector</i>)
Group 3	What challenges might occur in Cassandra–Hadoop integration (e.g., performance, consistency)?
Group 4	Suggest one innovative way to improve the integration for faster analytics.

CO – PO / PSO mapping:

(1 – Low 2 – Moderate 3 – High)

CO	PO1	PO3	PO4	PO9	PO11	PSO3
CO 4	3	2	2	1	1	2

PO / PSO mapped:

Innovative practice: CO4	PO1	PO3	PO4	PO9	PO11	PSO3
	3	2	2	1	1	2
Justification for correlation	Enables students to understand and utilize HDFS for storing and managing large-scale Big Data relevant to real-world analytics.	Encourages analysis of different storage and processing techniques (Cassandra + Hadoop) to solve complex Big Data problems.	Trains students to select and investigate appropriate data storage and processing techniques for diverse problem domains.	Promotes teamwork and collaborative decision-making while interpreting and integrating Big Data systems effectively.	Strengthens analytical, decision-making, and communication skills by interpreting large datasets and presenting conclusions to peers.	Develops ability to apply Big Data technologies to industrial and research problems for effective, real-world solutions.

Images / Screenshot of the practice :

Glimpse of Think Pair Share Activity



Figure1 and 2 shows the Think and Pair Timing in the class room

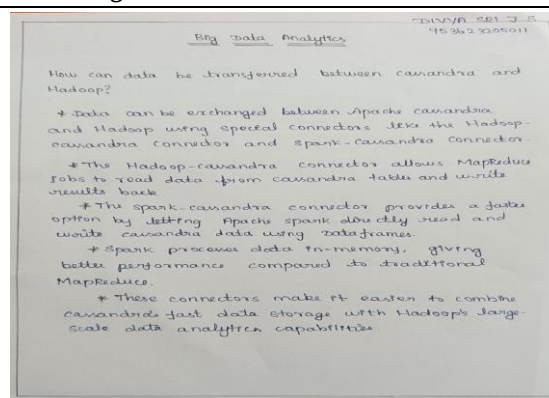
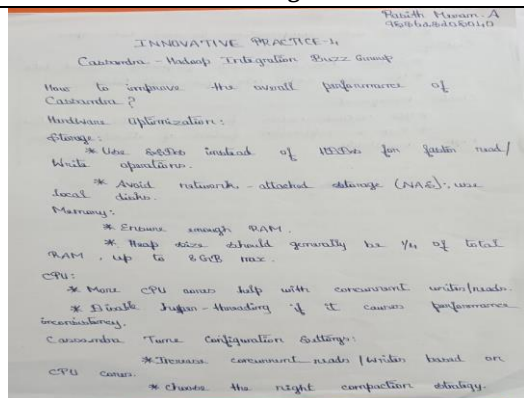


Figure3 and 4 shows the answers prepared by the students Rasith Meeran and Dhivya Sri for their group Questions

Feedback of practice from students and other stakeholders:

- Students found the **Buzz Group** session highly interactive and relevant to **Big Data integration practices**.
- Peer-to-peer learning helped in better understanding of **data flow and performance optimization** in Cassandra–Hadoop integration.
- Active discussions clarified **technical aspects** such as connector usage and consistency challenges.

❖ Benefit of the practice:

- Improved conceptual clarity on **Cassandra–Hadoop interoperability**.
- Enhanced **team participation** and classroom engagement.
- Students developed **confidence in explaining data analytics frameworks** used in industry.
- Encouraged **creative thinking** to propose innovative solutions for faster analytics.

❖ Challenges faced in implementation:

- A few students initially struggled to link technical concepts with real-world Big Data scenarios.
- Some groups required **additional guidance** to initiate idea exchange.
- Time limitation—an **extra 5 minutes** would have allowed more detailed presentation and peer questioning.

References:

- [Reading Rockets – Think-Pair-Share Strategy](#)
- [WGU – Think-Pair-Share Classroom Discussions](#)
- [Prodigy Education – Learning Strategies](#)

Signature of Faculty Member

HOD